Drummond B. Fielding

Curriculum Vitae

Assistant Professor
Department of Physics, New York University, Office 1053,
726 Broadway, New York, NY 10003
drummondfielding@gmail.com dfielding14.github.io arXiv ORCID

Education

PhD 2018, MA 2014, Astrophysics, University of California, Berkeley. *Advisor: Eliot Quataert*

BS, BA 2012, Physics, Mathematics, Johns Hopkins University.

Professional Appointments

Assistant Professor, New York University, Department of Physics, Center for Cosmology and Particle Physics	2025–
Associate Research Scientist, Flatiron Institute, Center for Computational Astrophysics	2025–
Assistant Professor, Cornell University, Department of Astronomy	2024–2025
Flatiron Research Fellow, CCA	2018–2024
Visiting Scholar, Indian Institute of Science	2016
NSF & Berkeley Graduate Research Fellow, U.C., Berkeley	2012–2018

Research interests

fluid dynamics	galactic winds	stellar+BH feedback
plasma physics	circumgalactic medium	magnetic reconnection
galaxy formation	magnetic turbulence	geom. measure theory
cosmic ray transport	multiphase flows	machine learning

Grants

The Geometry of Flows, Simons Foundation, PI, 2025–2030

Pushing the Frontier of Cosmic Ray Transport in Interstellar Turbulence, DOE INCITE, PI, 2025–2027

Resolving Cosmic Ray Transport by pushing the Frontier of MHD Turbulence, DOE INCITE, PI, 2024

Illuminating the Interplay of Multiphase Galactic Winds and the CGM, NSF AAG, PI

Mocking Galactic Winds: A new approach to constraining feedback by modeling the UV spectra of galaxies, Legacy HST-AR, Co-PI, 2025–2027

Galactic Winds Unveiled: Leveraging Cloud Simulations with Radiative Transfer to Constrain Feedback, Legacy HST-AR, Co-PI, 2024–2026

Resolving Turbulent Multiphase Gas Dynamics, EURO HPC, Co-PI, 2023–2025

AthenaK: a Performance Portable Simulation Infrastructure for Computational Astrophysics, NASA Support for Open-Source Tools, Frameworks, and Libraries, Co-I, 2025–2027

DUVET: Mapping Outflows and the Inner CGM in Starbursting Disks, KCWI, Co-I, 2020–

Visiting scholar grant to work Indian Institute of Science with Prof. Prateek Sharma, APS-IUSSTF, PI, 2016

Towards an Understanding of the Origin of OVI in the Circumgalactic Medium, HST AR, Co-I, 2017

The Physics of Supernova Feedback: Global 3D Simulations of Galactic Disks, NSF XSEDE, Co-I, 2016–2018

Conduction, Convection, and Thermal Instability in Hot Halos, NSF XSEDE, Co-I, 2016–2017

Advising

Brent Tan — Postdoc — Cornell University and NYU → Polymathic

Hitesh Das — Postdoc — NYU

Alex Pendris — Graduate Student — NYU

Maëlle Magnan — Graduate Student — NYU

Nana Ama Darpaah — Masters Student — NYU

Leela Elliott — Post-bac Student — NYU

Lillion Shi — Undergraduate — NYU

Sohum Udani — Undergraduate Student — NYU

Tiegan Powell — Undergraduate Student — NYU

Yongqi Zhang — Graduate Student — Cornell University

Bodong Liu — Undergraduate Student — Cornell University

Iryna Butsky — Graduate Student — University of Washington + CCA predoctoral fellow → Hubble Fellow at Stanford

Viraj Pandya — Graduate Student — UC Santa Cruz + CCA predoctoral fellow → Hubble Fellow at Columbia

Matthew Abruzzo — Graduate Student — Columbia → Postdoc at UPitt

Anthony Chow — Graduate Student — Columbia

Zirui Chen — Undergraduate Student — Columbia → Grad Student at UCSB

Minami Roy — Graduate Student — Raman Research Institute + CCA predoctoral fellow → Postdoc at OSU

Mitali Damle — Graduate Student — University of Potsdam → Postdoc at NYU Abu Dhabi

Professional services & teaching

Founder and organizer of the *New York Area Fluid Dynamics Meeting*, a cross-disciplinary monthly meeting of local fluid dynamicists, 2020

Research Mentor, *Simons-National Society of Black Physicists Scholars Program*, 2023

Flatiron Pre-Doctoral Program graduate student mentor, 2019–2024

Referee: MNRAS, ApJ, ApJL, PRL

Proposal evaluation panelist: NSF (2020), NASA HST (2024)

Honors and awards

Outstanding Graduate Student Instructor Award	2014
NSF Graduate Research Fellowship	2014–2017
Berkeley Graduate Fellowship	2012–2014
Donald E. Kerr Award for Outstanding Physics Undergraduate	2012
The Phi Beta Kappa Society	2012

Public outreach

Founder and leader of *The Astro Club*. Monthly meetings 2020–2023. Democracy Prep Charter Middle School, Harlem, NY.

Astronomy career day, Democracy Prep Charter Middle School, Harlem, NY, 2019

Taught a month-long series of astronomy classes to 2nd and 3rd grade students, 2017. North Oakland Community Charter School, Oakland, CA

UC Berkeley Astronomy Department Public Liaison, 2015

East Bay Astronomical Society Public Lecture, 2014, Chabot Space and Science Center

The Berkeley Compass Project Summer Evening Instructor, 2013

Selected recent presentations

Invited Seminar, Diffuse Gas in Cosmic Ecosystems Science Interest Group (7/2026)
Endowed Colloquium, Niels Bohr Lecture, Niels Bohr Institute, Copenhagen (10/2024)
Seminar, Institute for Advanced Studies, Princeton (10/2024)
Colloquium, Flatiron Institute, Center for Computational Astrophysics (2/2024)
Invited Review, *New simulations for new problems in galaxy formation*, conference, Paris (12/2023)
Invited Talk, Salpeter Workshop on the Interstellar Medium, Cornell (12/2023)
Seminar, Computational Research in Boston and Beyond (CRIBB), MIT (12/2023)
Colloquium, Johns Hopkins University, Physics and Astronomy Department (11/2023)
Colloquium, Columbia University, Astronomy Department (10/2023)
Invited talk, *Interfaces and Mixing in Fluids, Plasmas, and Materials*, conference, KITP (9/2023)
Colloquium, Michigan State University, Physics and Astronomy Department (10/2023)
Invited Review, *MIST2023: Cosmic turbulence and Magnetic fields: physics of baryonic matter across time and scales*, conference (9/2023)
Invited talk, *CGM@ND*, conference (9/2023)
Invited talk, *New Views on Feedback & the Baryon Cycle in Galaxies*, conference (8/2023)
Invited talk, *Modeling multiphase astrophysical systems*, conference (5/2023)
Invited talk, *Jim's Fest/Athena++*, conference (5/2023)
Colloquium, Yale University, Astronomy Department (2/2023)
Colloquium, Cornell University, Astronomy Department (1/2023)
Invited Astrophysics Seminar, UIUC (11/2022)
Joint Astrophysics Colloquium, Pitt-CMU (10/2022)
Invited talk, *What Matter(s) Around Galaxies*, conference (9/2022)
Invited talk, *Ringberg Computational Galaxy Formation 2022*, conference (4/2022)
Invited talk, *Wolfe Symposium*, conference (3/2022)
Invited Seminar, Theoretical AstroPhysics Including Relativity & Cosmology, Caltech (12/2021)
Astronomy Department seminar, Racah Institute of Physics, Hebrew University of Jerusalem (12/2021)
Invited Talk, *Baltimore Wind Workshop*, Johns Hopkins University (8/2021)
Astrophysics Colloquium, UCSD-SDSU (4/2021)
Keynote speaker, KITP workshop *Fundamentals of Gaseous Halos* (1/2021)
Invited talk, KITP conference *On the Origin, Nature, and Mixing of Multiphase Gas in Astrophysics* (10/2020)
Gaggle Seminar, Carnegie Institution for Science (8/2020)
Invited talk, CLEARNESS conference IAP (6/2020)
CGI Colloquium, U.C. Santa Cruz (4/2020)
Contributed Talk, Universality of Turbulence Conference, Flatiron Institute, NY (12/2019)
SFIR Seminar, Princeton University (11/2019)
Astro Seminar, NYU CCPP, New York, NY (10/2019)
Invited Talk, CGM conference, Berlin, Germany (10/2019)
Invited seminar, Turbulence workshop, Aspen Center for Physics (6/2019)
Invited Talk, *athena++* conference, Las Vegas, NV (3/2019)
Invited Talk, CGM conference, Northwestern, Chicago, IL (8/2018)
Invited Talk, CGM/DLA conference, Big Sur, CA (3/2018)
TAPIR Seminar, Caltech, Pasadena, CA (11/2017)
Astronomy Colloquium, Raman Research Institute (12/2016)
Astronomy Department Colloquium, Indian Institute of Science (10/2016)

Publications— *ADS search — Google Scholar — ORCID*

total publication count: 86 — citations: 4199 — h-index: 37 (*1 Nov 2026*)
* indicates $\geq$ 100 citations — ‡ graduate student-led — § undergraduate-led — † postdoc-led

First Author

8 **Fielding, Drummond B.**; Ripperda, Bart; Philippov, Alexander A., Plasmoid Instability in the Multiphase Interstellar Medium, ApJL, 949, L5, 2023 (arXiv:2211.06434) [46 citations]

*7 **Fielding, Drummond B.**; Bryan, Greg L., The Structure of Multiphase Galactic Winds, ApJ, 924, 82, 2022 (arXiv:2108.05355) [**177** citations]

6 **Fielding, Drummond B.**; Tonnesen, Stephanie; DeFelippis, Daniel; Li, Miao *et al.*, First Results from SMAUG: Uncovering the Origin of the Multiphase Circumgalactic Medium with a Comparative Analysis of Idealized and Cosmological Simulations, ApJ, 903, 32, 2020 (arXiv:2006.16316) [79 citations]

*5 **Fielding, Drummond B.**; Ostriker, Eve C.; Bryan, Greg L.; Jermyn, Adam S., Multiphase Gas and the Fractal Nature of Radiative Turbulent Mixing Layers, ApJL, 894, L24, 2020 (arXiv:2003.08390) [**216** citations]

*4 **Fielding, Drummond B.**; Quataert, Eliot; Martizzi, Davide, Clustered supernovae drive powerful galactic winds after superbubble breakout, MNRAS, 481, 3325, 2018 (arXiv:1807.08758) [**180** citations]

*3 **Fielding, Drummond B.**; Quataert, Eliot; Martizzi, Davide; Faucher-Giguère, Claude-André, How supernovae launch galactic winds?, MNRAS, 470, L39, 2017 (arXiv:1704.01579) [**100** citations]

*2 **Fielding, Drummond B.**; Quataert, Eliot; McCourt, Michael; Thompson, Todd A., The impact of star formation feedback on the circumgalactic medium, MNRAS, 466, 3810, 2017 (arXiv:1606.06734) [**193** citations]

*1 **Fielding, Drummond B.**; McKee, Christopher F.; Socrates, Aristotle; Cunningham, Andrew J. *et al.*, The turbulent origin of spin-orbit misalignment in planetary systems, MNRAS, 450, 3306, 2015 (arXiv:1409.5148) [**104** citations]

Second Author

21 Lancaster, Lachlan; **Fielding, Drummond B.**; Mohapatra, Rajsekhar; Bryan, Greg L., The Origin of Da Scaling: Suppressed Cooling in Fast-cooling Mixing Layers, ApJ, 1008, 13, 2026 (arXiv:2606.04093) [1 citations]

20 Xu, Xinfeng; **Fielding, Drummond B.**; Heckman, Timothy; Bryan, Greg L. *et al.*, Resolving the Unresolved Galactic Winds in Multiphase Models. I. Methodology and Application, ApJ, 1003, 155, 2026 (arXiv:2605.01105) [2 citations]

19 Kempski, Philipp; **Fielding, Drummond B.**; Quataert, Eliot; Ewart, Robert J. *et al.*, Self-similar Cosmic-Ray Transport in High-resolution Magnetohydrodynamic Turbulence, ApJL, 994, L49, 2025 (arXiv:2507.10651) [17 citations]

18 Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L.; Bennett, Jake S. *et al.*, ARKENSTONE - II. A model for unresolved cool clouds entrained in galactic winds in cosmological simulations, MNRAS, 535, 3550, 2024 (arXiv:2408.15321) [25 citations]

17 Stern, Jonathan; **Fielding, Drummond B.**; Hafen, Zachary; Su, Kung-Yi *et al.*, Correction to: accretion onto disc galaxies via hot and rotating CGM inflows, MNRAS, 532, 995, 2024

16 Stern, Jonathan; **Fielding, Drummond B.**; Hafen, Zachary; Su, Kung-Yi *et al.*, Accretion onto disc galaxies via hot and rotating CGM inflows, MNRAS, 530, 1711, 2024 (arXiv:2306.00092) [59 citations]

‡15 Abruzzo, Matthew W.; **Fielding, Drummond B.**; Bryan, Greg L., Taming the TuRMoiL: The Temperature Dependence of Turbulence in Cloud–Wind Interactions, ApJ, 966, 181, 2024 (arXiv:2210.15679) [39 citations]

†14 Tan, Brent; **Fielding, Drummond B.**, Cloud atlas: navigating the multiphase landscape of tempestuous galactic winds, MNRAS, 527, 9683, 2024 (arXiv:2305.14424) [63 citations]

13 Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L.; Kim, Chang-Goo *et al.*, ARKENSTONE - I. A novel method for robustly capturing high specific energy outflows in cosmological simulations, MNRAS, 527, 1216, 2024 (arXiv:2301.07116) [49 citations]

12 Kempski, Philipp; **Fielding, Drummond B.**; Quataert, Eliot; Galishnikova, Alisa K. *et al.*, Cosmic ray transport in large-amplitude turbulence with small-scale field reversals, MNRAS, 525, 4985, 2023 (arXiv:2304.12335) [92 citations]

‡11 Pandya, Viraj; **Fielding, Drummond B.**; Bryan, Greg L.; Carr, Christopher *et al.*, A Unified Model for the Coevolution of Galaxies and Their Circumgalactic Medium: The Relative Roles of Turbulence and Atomic Cooling Physics, ApJ, 956, 118, 2023 (arXiv:2211.09755) [56 citations]

‡10 Abruzzo, Matthew W.; **Fielding, Drummond B.**; Bryan, Greg L., TuRMoiL of Survival: A Unified Survival Criterion for Cloud-Wind Interactions, arXiv e-prints, 2023 (arXiv:2307.03228) [24 citations]

§9 Chen, Zirui; **Fielding, Drummond B.**; Bryan, Greg L., The Anatomy of a Turbulent Radiative Mixing Layer: Insights from an Analytic Model with Turbulent Conduction and Viscosity, ApJ, 950, 91, 2023 (arXiv:2211.01395) [30 citations]

‡8 Orr, Matthew E.; **Fielding, Drummond B.**; Hayward, Christopher C.; Burkhart, Blakesley, Bursting Bubbles: Feedback from Clustered Supernovae and the Trade-off Between Turbulence and Outflows, ApJ, 932, 88, 2022 (arXiv:2109.14656) [46 citations]

‡7 Orr, Matthew E.; **Fielding, Drummond B.**; Hayward, Christopher C.; Burkhart, Blakesley, Bursting Bubbles: Clustered Supernova Feedback in Local and High-redshift Galaxies, ApJL, 924, L28, 2022 (arXiv:2109.14626) [17 citations]

6 Stachenfeld, Kimberly; **Fielding, Drummond B.**; Kochkov, Dmitrii; Cranmer, Miles *et al.*, Learned Coarse Models for Efficient Turbulence Simulation, arXiv e-prints, 2021 (arXiv:2112.15275) [91 citations]

⋆‡5 Pandya, Viraj; **Fielding, Drummond B.**; Anglés-Alcázar, Daniel; Somerville, Rachel S. *et al.*, Characterizing mass, momentum, energy, and metal outflow rates of multiphase galactic winds in the FIRE-2 cosmological simulations, MNRAS, 508, 2979, 2021 (arXiv:2103.06891) [**179** citations]

⋆‡4 Butsky, Iryna S.; **Fielding, Drummond B.**; Hayward, Christopher C.; Hummels, Cameron B. *et al.*, The Impact of Cosmic Rays on Thermal Instability in the Circumgalactic Medium, ApJ, 903, 77, 2020 (arXiv:2008.04915) [**114** citations]

3 Stern, Jonathan; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André; Quataert, Eliot, The maximum accretion rate of hot gas in dark matter haloes, MNRAS, 492, 6042, 2020 (arXiv:1909.07402) [87 citations]

⋆2 Stern, Jonathan; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André; Quataert, Eliot, Cooling flow solutions for the circumgalactic medium, MNRAS, 488, 2549, 2019 (arXiv:1906.07737) [**113** citations]

⋆1 Martizzi, Davide; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André; Quataert, Eliot, Supernova feedback in a local vertically stratified medium: interstellar turbulence and galactic winds, MNRAS, 459, 2311, 2016 (arXiv:1601.03399) [**111** citations]

Co-author

57 Mohapatra, Rajsekhar; Lancaster, Lachlan; **Fielding, Drummond B.**; Quataert, Eliot *et al.*, How Magnetic Fields Regulate Cooling and Mixing in Turbulent Radiative Mixing Layers, arXiv e-prints, 2026 (arXiv:2609.29782)

56 Sullivan, James M.; Bryan, Greg L.; Smith, Matthew C.; Bennett, Jake S. *et al.* (incl. **DBF**), ARKENSTONE BH: a model for high specific energy black hole feedback in cosmological simulations, MNRAS, 551, stag1520, 2026 (arXiv:2605.03154) [3 citations]

55 Lin, Allison; Lancaster, Lachlan; Pandya, Viraj; **Fielding, Drummond B.** *et al.*, Illuminating M82: Simulating X-ray Emission from Galactic Winds in a Starburst Galaxy, arXiv e-prints, 2026 (arXiv:2607.12087)

54 Sankar, Sriram; Stern, Jonathan; Power, Chris; Catinella, Barbara *et al.* (incl. **DBF**), Hot accretion onto spiral galaxies: the origin of extended and warped H I discs, MNRAS, 549, stag1059, 2026 (arXiv:2511.03793) [9 citations]

53 Lancaster, Lachlan; Mohapatra, Rajsekhar; **Fielding, Drummond B.**; Bryan, Greg L., Ceci n'est pas une Couche de Mélange: The Meaning of Resolved Turbulent Radiative Mixing, ApJ, 1005, 222, 2026 (arXiv:2606.04087) [2 citations]

52 Butsky, Iryna S.; Redshaw, Caleb; Lei, Minjie; Clark, Susan E. *et al.* (incl. **DBF**), Phase-dependent magnetic coherence in the turbulent interstellar medium, arXiv e-prints, 2026 (arXiv:2606.11323) [2 citations]

51 Chen, Zirui; Oh, S. Peng; **Fielding, Drummond B.**; Lancaster, Lachlan *et al.*, Clumps in a Cocoon: Geometry and Mixing Set the Universal X-ray to Hα Surface Brightness Ratio, arXiv e-prints, 2026 (arXiv:2606.07741) [3 citations]

50 Goldner, Roy; Stern, Jonathan; **Fielding, Drummond B.**; Faucher-Giguère, Claude-André *et al.*, Accretion-driven turbulence in the circumgalactic medium, MNRAS, 548, stag717, 2026 (arXiv:2510.27678) [11 citations]

49 Chen, Zirui; Peng, Zixuan; Rubin, Kate H. R.; Heckman, Timothy M. *et al.* (incl. **DBF**), Modelling emission-line surface brightness in a multiphase galactic wind: an O VI case study, MNRAS, 548, stag617, 2026 (arXiv:2510.02443) [3 citations]

48 Pandya, Viraj; Bryan, Greg L.; Makinen, T. Lucas; Gabrielpillai, Austen *et al.* (incl. **DBF**), Introducing sapphire: Towards Hybrid Physics-Informed, Data-Driven Modeling of Galaxy Formation, arXiv e-prints, 2026 (arXiv:2604.06318) [6 citations]

47 Bennett, Jake S.; Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L. *et al.*, Correction to: Prevention is better than cure? Feedback from high specific energy winds in cosmological simulations with ARKENSTONE, MNRAS, 547, stag374, 2026 [2 citations]

⋆46 Stone, James M.; Mullen, Patrick D.; **Fielding, Drummond B.**; Grete, Philipp *et al.*, AthenaK: A Performance-portable Version of the Athena++ Adaptive Mesh Refinement Framework, ApJS, 283, 27, 2026 (arXiv:2409.16053) [**115** citations]

45 Ramesh, Rahul; Nelson, Dylan; **Fielding, Drummond B.**; Brüggen, Marcus, Zooming in on the circumgalactic medium with GIBLE: Cloud-scale simulations with cosmological initial conditions, arXiv e-prints, 2026 (arXiv:2602.23416) [9 citations]

44 Burger, Jan D.; Springel, Volker; Ostriker, Eve C.; Kim, Chang-Goo *et al.* (incl. **DBF**), Applying a star formation model calibrated on high-resolution interstellar medium simulations to cosmological simulations of galaxy formation, MNRAS, 544, 1390, 2025 (arXiv:2502.13244) [13 citations]

43 Kakoly, Aharon; Stern, Jonathan; Faucher-Giguère, Claude-André; **Fielding, Drummond B.** *et al.*, Turbulence-dominated CGM: the origin of UV absorbers with equivalent widths of $\sim$1 Å, MNRAS, 543, 3345, 2025 (arXiv:2504.17001) [21 citations]

42 Lei, Minjie; Clark, S. E.; Morel, Rudy; Allys, E. *et al.* (incl. **DBF**), Neutral Gas Phase Distribution from H I Morphology: Phase Separation with Scattering Spectra and Variational Autoencoders, ApJ, 993, 4, 2025 (arXiv:2505.20407) [6 citations]

41 Bennett, Jake S.; Smith, Matthew C.; **Fielding, Drummond B.**; Bryan, Greg L. *et al.*, Prevention is better than cure? Feedback from high specific energy winds in cosmological simulations with ARKENSTONE, MNRAS, 543, 1456, 2025 (arXiv:2410.12909) [20 citations]

40 Porter, Lori E.; Abruzzo, Matthew; Bryan, Greg L.; Putman, Mary *et al.* (incl. **DBF**), Simulating high-velocity clouds in the observational plane: an initial study with the Smith Cloud, MNRAS, 542, 1188, 2025 (arXiv:2506.00111) [1 citations]

39 Steinwandel, Ulrich P.; Rennehan, Douglas; Orr, Matthew E.; **Fielding, Drummond B.** *et al.*, Pumping Iron: How Turbulent Metal Diffusion Impacts Multiphase Galactic Outflows, ApJ, 991, 16, 2025 (arXiv:2407.14599) [13 citations]

38 Kempski, Philipp; Li, Dongzi; **Fielding, Drummond B.**; Quataert, Eliot *et al.*, A Unified Model of Cosmic-Ray Propagation and Radio Extreme Scattering Events from Intermittent Interstellar Structures, ApJL, 990, L18, 2025 (arXiv:2412.03649) [19 citations]

37 Mohapatra, Rajsekhar; Quataert, Eliot; **Fielding, Drummond B.**; Guo, Minghao, The Type Ia Supernova and Asymptotic Giant Branch Stellar Ejecta-regulated Interstellar Medium of Massive Galaxies, ApJ, 989, 103, 2025 (arXiv:2502.05329) [5 citations]

36 Ramesh, Rahul; Nelson, Dylan; **Fielding, Drummond B.**; Brüggen, Marcus, Zooming in on the circumgalactic medium with GIBLE: Tracing the origin and evolution of cold clouds, A&A, 699, A10, 2025 (arXiv:2407.00172) [18 citations]

35 Fisher, Deanne B.; Bolatto, Alberto D.; Chisholm, John; **Fielding, Drummond B.** *et al.*, JWST observations of starbursts: cold clouds and plumes launching in the M 82 outflow, MNRAS, 538, 3068, 2025 (arXiv:2405.03686) [32 citations]

34 Peng, Zixuan; Martin, Crystal L.; Chen, Zirui; **Fielding, Drummond B.** *et al.*, Physical Origins of Outflowing Cold Clouds in Local Star-forming Dwarf Galaxies, ApJ, 981, 171, 2025 (arXiv:2412.05371) [13 citations]

33 Reichardt Chu, Bronwyn; Fisher, Deanne B.; Chisholm, John; Berg, Danielle *et al.* (incl. **DBF**), DUVET: sub-kiloparsec resolved star formation driven outflows in a sample of local starbursting disc galaxies, MNRAS, 536, 1799, 2025 (arXiv:2402.17830) [32 citations]

32 Qutob, Nadia; Emami, Razieh; Su, Kung-Yi; Smith, Randall *et al.* (incl. **DBF**), Observational Signatures of AGN Feedback in the Morphology and the Ionization States of Milky Way-like Galaxies, ApJ, 977, 72, 2024 (arXiv:2312.14809) [5 citations]

31 Choi, Bo-Eun; Werk, Jessica K.; Tchernyshyov, Kirill; Prochaska, J. Xavier *et al.* (incl. **DBF**), Metallicity Mapping of the Ionized Diffuse Gas at the Milky Way Disk–Halo Interface, ApJ, 976, 222, 2024 (arXiv:2410.06286) [3 citations]

30 Voit, G. Mark; Carr, Christopher; **Fielding, Drummond B.**; Pandya, Viraj *et al.*, Equilibrium States of Galactic Atmospheres. II. Interpretation and Implications, ApJ, 976, 151, 2024 (arXiv:2406.07632) [18 citations]

29 Voit, G. Mark; Pandya, Viraj; **Fielding, Drummond B.**; Bryan, Greg L. *et al.*, Equilibrium States of Galactic Atmospheres. I. The Flip Side of Mass Loading, ApJ, 976, 150, 2024 (arXiv:2406.07631) [23 citations]

28 Ohana, Ruben; McCabe, Michael; Meyer, Lucas; Morel, Rudy *et al.* (incl. **DBF**), The Well: a Large-Scale Collection of Diverse Physics Simulations for Machine Learning, arXiv e-prints, 2024 (arXiv:2412.00568) [27 citations]

27 Perrotta, Serena; Coil, Alison L.; Rupke, David S. N.; Ning, Wenmeng *et al.* (incl. **DBF**), The Outflowing [O II] Nebulae of Compact Starburst Galaxies at z $\sim$ 0.5, ApJ, 975, 263, 2024 (arXiv:2409.10013) [11 citations]

26 Hassan, Sultan; Ostriker, Eve C.; Kim, Chang-Goo; Bryan, Greg L. *et al.* (incl. **DBF**), Toward Implementation of the Pressure-regulated, Feedback-modulated Model of Star Formation in Cosmological Simulations: Methods and Application to TNG, ApJ, 975, 151, 2024 (arXiv:2409.09121) [12 citations]

25 Hummels, Cameron B.; Rubin, Kate H. R.; Schneider, Evan E.; **Fielding, Drummond B.**, CLOUDFLEX: A Flexible Parametric Model for the Small-scale Structure of the Circumgalactic Medium, ApJ, 972, 148, 2024 (arXiv:2311.05691) [27 citations]

24 Kim, Chang-Goo; Ostriker, Eve C.; Kim, Jeong-Gyu; Gong, Munan *et al.* (incl. **DBF**), Metallicity Dependence of Pressure-

regulated Feedback-modulated Star Formation in the TIGRESS-NCR Simulation Suite, ApJ, 972, 67, 2024 (arXiv:2405.19227) [25 citations]

23 Su, Kung-Yi; Bryan, Greg L.; Hayward, Christopher C.; Somerville, Rachel S. *et al.* (incl. **DBF**), Unravelling jet quenching criteria across L* galaxies and massive cluster ellipticals, MNRAS, 532, 2724, 2024 (arXiv:2310.17692) [24 citations]

22 Hamel-Bravo, Magdalena J.; Fisher, Deanne B.; Berg, Danielle; Björgvinsson, Bjarki *et al.* (incl. **DBF**), DUVET: Resolved direct metallicity measurements in the outflow of starburst galaxy NGC 1569, MNRAS, 530, 3855, 2024 (arXiv:2404.04600) [20 citations]

‡21 Ramesh, Rahul; Nelson, Dylan; **Fielding, Drummond B.**; Brüggen, Marcus, Zooming in on the circumgalactic medium with GIBLE. The topology and draping of magnetic fields around cold clouds, A&A, 684, L16, 2024 (arXiv:2404.01370) [25 citations]

20 Coil, Alison L.; Perrotta, Serena; Rupke, David S. N.; Lochhaas, Cassandra *et al.* (incl. **DBF**), Ionized gas extends over 40 kpc in an odd radio circle host galaxy, Nature, 625, 459, 2024 (arXiv:2310.15162) [27 citations]

19 Roy, Manami; Su, Kung-Yi; Tonnesen, Stephanie; **Fielding, Drummond B.** *et al.*, Seeding the CGM: how satellites populate the cold phase of milky way haloes, MNRAS, 527, 265, 2024 (arXiv:2310.04404) [23 citations]

18 Steinwandel, Ulrich P.; Kim, Chang-Goo; Bryan, Greg L.; Ostriker, Eve C. *et al.* (incl. **DBF**), The Structure and Composition of Multiphase Galactic Winds in a Large Magellanic Cloud Mass Simulated Galaxy, ApJ, 960, 100, 2024 (arXiv:2212.03898) [58 citations]

17 McPherson, Daniel K.; Fisher, Deanne B.; Nielsen, Nikole M.; Kacprzak, Glenn G. *et al.* (incl. **DBF**), DUVET survey: mapping outflows in the metal-poor starburst Mrk 1486, MNRAS, 525, 6170, 2023 (arXiv:2308.06918) [17 citations]

‡16 Carr, Christopher; Bryan, Greg L.; **Fielding, Drummond B.**; Pandya, Viraj *et al.*, Regulation of Star Formation by a Hot Circumgalactic Medium, ApJ, 949, 21, 2023 (arXiv:2211.05115) [48 citations]

15 Reichardt Chu, Bronwyn; Fisher, Deanne B.; Bolatto, Alberto D.; Chisholm, John *et al.* (incl. **DBF**), DUVET: Spatially Resolved Observations of Star Formation Regulation via Galactic Outflows in a Starbursting Disk Galaxy, ApJ, 941, 163, 2022 (arXiv:2211.02063) [17 citations]

⋆14 Hafen, Zachary; Stern, Jonathan; Bullock, James; Gurvich, Alexander B. *et al.* (incl. **DBF**), Hot-mode accretion and the physics of thin-disc galaxy formation, MNRAS, 514, 5056, 2022 (arXiv:2201.07235) [**122** citations]

13 Butsky, Iryna S.; Werk, Jessica K.; Tchernyshyov, Kirill; **Fielding, Drummond B.** *et al.*, The Impact of Cosmic Rays on the Kinematics of the Circumgalactic Medium, ApJ, 935, 69, 2022 (arXiv:2106.14889) [38 citations]

‡12 Abruzzo, Matthew W.; Bryan, Greg L.; **Fielding, Drummond B.**, A Simple Model for Mixing and Cooling in Cloud-Wind Interactions, ApJ, 925, 199, 2022 (arXiv:2101.10344) [68 citations]

11 Stern, Jonathan; Sternberg, Amiel; Faucher-Giguère, Claude-André; Hafen, Zachary *et al.* (incl. **DBF**), Neutral CGM as damped Ly α absorbers at high redshift, MNRAS, 507, 2869, 2021 (arXiv:2105.06489) [45 citations]

⋆10 Stern, Jonathan; Faucher-Giguère, Claude-André; **Fielding, Drummond B.**; Quataert, Eliot *et al.*, Virialization of the Inner CGM in the FIRE Simulations and Implications for Galaxy Disks, Star Formation, and Feedback, ApJ, 911, 88, 2021 (arXiv:2006.13976) [**158** citations]

9 Burkhart, B.; Appel, S. M.; Bialy, S.; Cho, J. *et al.* (incl. **DBF**), The Catalogue for Astrophysical Turbulence Simulations (CATS), ApJ, 905, 14, 2020 (arXiv:2010.11227) [36 citations]

8 Pandya, Viraj; Somerville, Rachel S.; Anglés-Alcázar, Daniel; Hayward, Christopher C. *et al.* (incl. **DBF**), First Results from SMAUG: The Need for Preventative Stellar Feedback and Improved Baryon Cycling in Semianalytic Models of Galaxy Formation, ApJ, 905, 4, 2020 (arXiv:2006.16317) [58 citations]

7 Kim, Chang-Goo; Ostriker, Eve C.; **Fielding, Drummond B.**; Smith, Matthew C. *et al.*, A Framework for Multiphase Galactic Wind Launching Using TIGRESS, ApJL, 903, L34, 2020 (arXiv:2010.09090) [72 citations]

⋆6 Kim, Chang-Goo; Ostriker, Eve C.; Somerville, Rachel S.; Bryan, Greg L. *et al.* (incl. **DBF**), First Results from SMAUG: Characterization of Multiphase Galactic Outflows from a Suite of Local Star-forming Galactic Disk Simulations, ApJ, 900, 61, 2020 (arXiv:2006.16315) [**157** citations]

5 Lochhaas, Cassandra; Bryan, Greg L.; Li, Yuan; Li, Miao *et al.* (incl. **DBF**), Properties of the simulated circumgalactic medium, MNRAS, 493, 1461, 2020 (arXiv:1908.00021) [48 citations]

4 Martizzi, Davide; Quataert, Eliot; Faucher-Giguère, Claude-André; **Fielding, Drummond B.**, Simulations of jet heating in galaxy clusters: successes and challenges, MNRAS, 483, 2465, 2019 (arXiv:1805.06461) [65 citations]

3 Stern, Jonathan; Faucher-Giguère, Claude-André; Hennawi, Joseph F.; Hafen, Zachary *et al.* (incl. **DBF**), Does Circumgalactic O VI Trace Low-pressure Gas Beyond the Accretion Shock? Clues from H I and Low-ion Absorption, Line Kinematics, and Dust

Extinction, ApJ, 865, 91, 2018 (arXiv:1803.05446) [59 citations]

⋆2 Offner, Stella S. R.; Dunham, Michael M.; Lee, Katherine I.; Arce, Héctor G. *et al.* (incl. **DBF**), The Turbulent Origin of Outflow and Spin Misalignment in Multiple Star Systems, ApJL, 827, L11, 2016 (arXiv:1606.08445) [**126** citations]

1 Schlieder, Joshua E.; Lépine, Sébastien; Rice, Emily; Simon, Michal *et al.* (incl. **DBF**), The Na 8200 Å Doublet as an Age Indicator in Low-mass Stars, AJ, 143, 114, 2012 (arXiv:1202.4191) [69 citations]